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Exponentials for Free

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Abstract

Functional programming languages modulo $\beta\eta$-equations can be understood as freely generated categorical structures [1]. This classical correspondence provides a method for constructing program transformations: specify the images of primitive types and operations, and let the universal property determine the transformation. In a suitable logical-relations model, correctness likewise extends from the primitives to all programs, yielding transformations that are correct by construction.

This perspective has proved fruitful in automatic differentiation for ML-family languages [2] and in CHAD for expressive total languages [3], with developments addressing iteration and non-termination [4], simply typed implementations [5], and systematic optimisation [6]. It also points towards further extensions to probabilistic languages [7] and the compilation of value-indexed analyses beyond differentiation [8].

It is therefore natural to explore the same semantics-driven approach to defunctionalisation. We approach this problem by studying the existence of exponentials and their weak variants in freely generated categorical structures. The talk focuses on this question. I will discuss how function spaces emerge from free constructions, building on results on free extensivity and doubly-infinitary distributive categories [9, 10]. The emphasis will be on the categorical ideas and their universal properties.

The talk draws on several strands of work, including joint work with Rui Prezado and Matthijs Vákár [9], and is based primarily on ongoing joint work with Matthijs Vákár [11].

Keywords: Multicategories; free completions; cartesian closedness; defunctionalisation

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